

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10: SUB-STRAND 2.7 SURFACE AREA AND VOLUME OF SOLIDS – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why does the same amount of metal sheet make containers that hold very different amounts of grain, and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction? Use what you learned in Lessons 1 to 8 to answer each question below. Write in full sentences, show every step of your calculations, include units, and refer to the posho mill in Eldoret where you can. Round answers to the accuracy asked for. Use your Driving Question Board and your cumulative model portfolio to help you.

Part 1: What the metal sheet really tells us

A posho mill owner in Eldoret has a rectangular metal box for storing maize. It measures $l = 120$ cm, $w = 80$ cm and $h = 100$ cm. The fundi who made it used the same total area of iron sheet for a cylindrical drum. (a) Write down the formula you use and calculate the total area of iron sheet used for the box. (b) Calculate the volume of the box in cm^3 and in m^3 . (c) In your own words, explain what surface area measures and what volume measures. Say why knowing one of them does not tell you the other.

Write your answer here:

Part 2: Solving the posho mill mystery with numbers

The cylindrical drum is closed at top and bottom, is 100 cm tall, and is made from the same 59,200 cm² of iron sheet as the box. Its radius works out to about 59.2 cm. (a) Show, using the formula for the total surface area of a closed cylinder, that this radius matches the sheet area given. A small rounding difference is acceptable. (b) Calculate the volume of the drum in cm³ and in m³ to 2 decimal places. (c) Compare it with the box from Part 1. State which holds more maize, by how many cm³, and by about what percentage of the box's volume.

Write your answer here:

Part 3: Why the cylinder wins

Use your understanding of shapes to explain to the mill owner why the cylindrical drum holds more maize than the rectangular box even though both used the same area of iron sheet. In your explanation: (a) say what the same-height drum and box have in common and what is different about their base shapes; (b) use the idea of the isoperimetric principle from Lesson 8; (c) explain why this means your answer is not just a coincidence of these particular dimensions.

Write your answer here:

Part 4: Choosing the right formula for the fundi

A metalsmith makes a conical grain funnel with base radius $r = 30$ cm and vertical height $h = 40$ cm. (a) Calculate the slant height. Explain why the slant height, not the vertical height, is used to find the sheet metal needed. (b) Find the curved surface area of the funnel, leaving your answer in terms of π and then to the nearest cm^2 . (c) Find the volume of the cone in terms of π and then to the nearest cm^3 . (d) State how the volume of this cone compares with a cylinder of the same base radius and height, and how you know.

Write your answer here:

Part 5: Advice for a Kenyan artisan

A jua kali metalsmith in Kamukunji has a fixed amount of iron sheet and wants to build a grain store that holds as much maize as possible. Write a short piece of advice to the metalsmith that answers the driving question. Your advice must: (a) explain the difference between surface area and volume and say how the metalsmith can calculate each for a cylinder and for a cuboid; (b) recommend a shape from the Eldoret example and justify it with evidence from your calculations; (c) explain what the metalsmith should do for a composite solid such as a cylinder topped with a cone or hemisphere, including how to treat surfaces where two parts join.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Correct use of surface area and volume formulae and calculations	Chooses the right formula every time, shows all steps, gets correct values with units and sensible rounding for the box, drum and cone.	Uses mostly correct formulae and gets most values right, with minor arithmetic, rounding or unit slips.	Uses wrong formulae or leaves out steps, with several incorrect values or missing units.
Distinguishing surface area from volume	Clearly explains that surface area is the material covering a solid and volume is the space inside, and uses the Eldoret containers to show they are independent.	States the difference correctly but gives a limited link to the phenomenon.	Confuses surface area and volume or treats equal surface area as meaning equal volume.
Explaining why the cylinder holds more (isoperimetric reasoning)	Links the circular cross-section to a greater base area for the same perimeter and explains that this applies generally, not just to these numbers.	Says the circle is more efficient and links this to volume, but the reasoning is incomplete.	Gives no mathematical reason, or only says the cylinder is bigger.
Use of evidence and Kenyan context in the design advice	Uses specific data from the Eldoret example, handles slant height and composite solids correctly, and gives practical, justified advice.	Uses some evidence and gives reasonable advice, with small gaps on slant height or composite solids.	Gives advice with little or no evidence, or omits the composite solid or slant height ideas.

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